

Chapter 6

Basic SQL 2

1

Ambiguous Attribute Names

Since same name can be used for attributes in different relations, we must **qualify** the attribute name with the relation name to prevent ambiguity

```
SELECT FName, LName, Address
FROM EMPLOYEE, DEPARTMENT
WHERE Dname='Research' AND Dnumber=Dno;
```

```
SELECT FName, EMPLOYEE.LName, Address
FROM EMPLOYEE, DEPARTMENT
WHERE DEPARTMENT.Name='Research' AND
      DEPARTMENT.Dnumber=EMPLOYEE.Dnumber;
```

2

Aliasing or Renaming Tuple Variables

Query: For each employee, retrieve the employee's first and last name and the first and last name of his or her immediate supervisor.

```
SELECT E.FName, E.LName, S.Fname AS S_Fname, S.Lname AS S_Lname
FROM   EMPLOYEE E, EMPLOYEE S
WHERE  E.Super_ssn=S.Ssn;
```

Fname	Lname	S_Fname	S_Lname
John	Smith	Franklin	Wong
Franklin	Wong	James	Borg
Alicia	Zalaya	Jennifer	Wallace
Jennifer	Wallace	James	Borg
Ramesh	Narayan	Franklin	Wong
Joyce	English	Franklin	Wong
Ahmad	Jabbar	Jennifer	Wallace

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Unspecified WHERE Clause

- Missing **WHERE** clause
 - Indicates no condition on tuple selection
- Effect is a **CARTESIAN PRODUCT**
 - Result is all possible tuple combinations

Query: Select all **EMPLOYEE** SSN's

```
SELECT Ssn
FROM EMPLOYEE;
```

Ssn
123456789
333445555
453453453
666884444
888665555
987654321
987987987
999887777

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Query: Select all combinations of **EMPLOYEE** Ssn and **DEPARTMENT** Dname in the database.

```
SELECT  Ssn, Dname
FROM    EMPLOYEE, DEPARTMENT;
```

Ssn	Dname
123456789	Administration
333445555	Administration
453453453	Administration
666884444	Administration
888665555	Administration
987654321	Administration
987987987	Administration
999887777	Administration
123456789	Headquarters
333445555	Headquarters
453453453	Headquarters
666884444	Headquarters
888665555	Headquarters
987654321	Headquarters
987987987	Headquarters
999887777	Headquarters
123456789	Research
333445555	Research
453453453	Research
666884444	Research
888665555	Research
987654321	Research
987987987	Research
999887777	Research

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The Asterisk

An asterisk (*) may be used to retrieve all the attribute values of the selected tuples

The * can be prefixed by the relation name; e.g., **EMPLOYEE.***

```
SELECT *
FROM EMPLOYEE
WHERE Dno=5;
```

```
SELECT *
FROM EMPLOYEE,DEPARTMENT;
```

```
SELECT EMPLOYEE.*
FROM EMPLOYEE,DEPARTMENT
WHERE Dno=Dnumber AND Dname='Research';
```

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Tables as Sets

- SQL does not automatically eliminate duplicate tuples in query results
- For aggregate operations duplicates must be accounted for
- Use the keyword **DISTINCT** in the **SELECT** clause to eliminate duplicates

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Query: Retrieve the salary of every employee.

```
SELECT Salary  
FROM EMPLOYEE;
```

Salary
30000
40000
25000
43000
38000
25000
25000
55000

Query: Retrieve all distinct salary values.

```
SELECT DISTINCT Salary  
FROM EMPLOYEE;
```

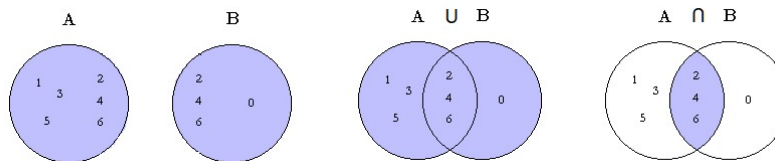
Salary
30000
40000
25000
43000
38000
55000

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Tables as Sets

Set operations:

- **UNION, EXCEPT** (difference), **INTERSECT**
- Corresponding *multiset* operations: **UNION ALL, EXCEPT ALL, INTERSECT ALL**)
- Type compatibility is required for these operations to be valid



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Tables as Sets

Query: List all project numbers for projects that involve an employee whose last name is 'Wong', either as a **worker** or as a manager of the department that controls the project.

Join of two tables

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

Pnumber
2
3
10
20

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

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Tables as Sets

Query: List all project numbers for projects that involve an employee whose last name is 'Wong', either as a worker or as a **manager** of the department that controls the project.

Join of three tables

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

Pnumber
1
2
3

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Tables as Sets in SQL

Query: List all project numbers for projects that involve an employee whose last name is 'Wong', either as a worker or as a manager of the department that controls the project.

```
SELECT DISTINCT Pno
  FROM WORKS_ON, EMPLOYEE
 WHERE Essn=Ssn AND Lname='Wong'

UNION

SELECT DISTINCT PNUMBER
  FROM PROJECT, DEPARTMENT, EMPLOYEE
 WHERE Dnum=Dnumber AND Mgr_ssn=Ssn AND Lname='Wong';
```

Pnumber
2
3
10
20

Pnumber
1
2
3
10
20

Pnumber
1
2
3

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Substring Pattern Matching

LIKE comparison operator

- used for string **pattern matching**
- % replaces an arbitrary number of zero or more characters
- underscore (_) replaces a single character

```
WHERE Address LIKE '%Houston TX%';
```

```
WHERE Ssn LIKE '__ 1__ 8901';
```

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Arithmetic Operations

Standard arithmetic operators:

- Addition (+), subtraction (–), multiplication (*), and division (/) may be included as a part of **SELECT**

Query: Show the resulting salaries if every employee working on the 'ProductX' project is given a 10% raise.

Join of three tables

```
SELECT FName, LName, 1.1*Salary AS NewSalary  
FROM   EMPLOYEE, WORKS_ON, PROJECT  
WHERE  Pname='ProductX' AND Pno=Pnumber AND Ssn=Essn;
```

Fname	Lname	NewSalary
John	Smith	33000
Joyce	English	27500

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Arithmetic Operators

BETWEEN comparison operator:

```
WHERE (Salary BETWEEN 30000 AND 40000) AND Dno = 5;
```

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Ordering of Query Results

Use **ORDER BY** clause

- Use **ASC** or **DESC** to order result in ascending or descending order
- Typically placed at the end of the query

Query: Show the salaries in the company which are higher than \$27,000, in increasing order.

```
SELECT DISTINCT Salary  
FROM EMPLOYEE  
WHERE Salary > 27000  
ORDER BY Salary;
```

Salary
30000
38000
40000
43000
55000

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Modifying the Database

- **INSERT** inserts new tuples in a relation
- **UPDATE** modifies tuples that satisfy the condition in a relation
- **DELETE** removes tuples that satisfy the condition from a relation

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INSERT

- In its simplest form, used to add one or more tuples to a relation.
- Attribute values should be listed in the same order specified in the **CREATE TABLE** command.
- Constraints on data types are enforced.
- All integrity constraints are enforced.

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INSERT

Specify the relation name and a list of values for the tuple. All values including nulls must be supplied.

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
-------	-------	-------	-----	-------	---------	-----	--------	-----------	-----

```

INSERT INTO EMPLOYEE VALUES
('John', 'B', 'Smith', '123456789', '1965-01-09', '731 Fondren, Houston TX', 'M', 30000, '333445555', 5);

INSERT INTO DEPARTMENT VALUES ('Research', 5, '333445555', '1988-05-22');

INSERT INTO PROJECT VALUES ('ProductX', 1, 'Bellaire', 5);

INSERT INTO WORKS_ON VALUES ('123456789', 1, 32.5);

INSERT INTO DEPENDENT VALUES ('333445555', 'Alice', 'F', '1986-04-04', 'Daughter');

INSERT INTO DEPT_LOCATIONS VALUES (1, 'Houston');
    
```

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INSERT

Example: A new table is created and populated by the result of a query. The table contains employee last name, project name and hours per week for each employee working on some project.

```

CREATE TABLE WORKS_ON_INFO (
    Emp_name      VARCHAR(15),
    Proj_name     VARCHAR(15),
    Hours_per_week DECIMAL(3,1)
);

INSERT INTO WORKS_ON_INFO
SELECT Lname, Pname, Hours
FROM PROJECT, WORKS_ON, EMPLOYEE
WHERE Pnumber=Pno AND Essn=Ssn;
    
```

WORKS_ON_INFO

Emp_name	Proj_name	Hours_per_week
Smith	ProductX	32.5
Smith	ProductY	7.5
Narayan	ProductZ	40
English	ProductX	20
English	ProductY	20
Wong	ProductY	10
Wong	ProductZ	10
Wong	Computerization	10
Wong	Reorganization	10
Zalaya	Newbenefits	30
Zalaya	Computerization	10
Jabbar	Computerization	35
Jabbar	Newbenefits	5
Wallace	Newbenefits	20
Wallace	Reorganization	15
Borg	Reorganization	NULL

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INSERT

Example: A new table is created and populated in one shot by the result of a query.

```
CREATE TABLE WORKS_ON_INFO AS
SELECT Lname,Pname,Hours
FROM PROJECT, WORKS_ON, EMPLOYEE
WHERE Pnumber=Pno AND Essn=Ssn;
```

WORKS_ON_INFO

Emp_name	Proj_name	Hours_per_week
Smith	ProductX	32.5
Smith	ProductY	7.5
Narayan	ProductZ	40
English	ProductX	20
English	ProductY	20
Wong	ProductY	10
Wong	ProductZ	10
Wong	Computerization	10
Wong	Reorganization	10
Zalaya	Newbenefits	30
Zalaya	Computerization	10
Jabbar	Computerization	35
Jabbar	Newbenefits	5
Wallace	Newbenefits	20
Wallace	Reorganization	15
Borg	Reorganization	NULL

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Bulk Loading of Tables

Example: A new table of the employees of department #5 is created and populated by the result of a query.

```
CREATE TABLE D5EMPS AS
SELECT *
FROM EMPLOYEE
WHERE Dno=5 ;
```

D5EMPS

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1/9/1965	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	12/8/1955	638 Voss, Houston, TX	M	40000	888665555	5
Ramesh	K	Narayan	666884444	9/15/1962	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	7/31/1972	5631 Rice, Houston, TX	F	25000	333445555	5

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DELETE

Removes tuples from a relation.

The **WHERE** clause specifies the tuples to be deleted.

```
DELETE FROM DEPENDENT
WHERE Dependent_name='Joy' OR Relationship='Spouse';
```

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	04/05/1986	Daughter
333445555	Theodore	M	10/25/1983	Son
123456789	Michael	M	01/04/1988	Son
123456789	Alice	F	12/30/1988	Daughter

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DELETE

- Removes tuples from a relation
- Referential integrity should be enforced
- Tuples are deleted from only **one table** at a time (unless **CASCADE** is specified on a referential integrity constraint)
- A missing **WHERE** clause specifies that **all tuples** in the relation are to be deleted; the table then becomes an empty table

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UPDATE

- Modifies attribute values of one or more selected tuples
- A **WHERE** clause selects the tuples to be modified
- An additional **SET** clause specifies the attributes to be modified and their new values
- Each command modifies tuples **in the same relation**
- Referential integrity is enforced

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UPDATE

Example: Change the location and controlling department number of project number 10 to 'Bellaire' and 5, respectively.

```
UPDATE PROJECT
SET Plocation='Bellaire',Dnum=5
WHERE Pnumber=10;
```

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4



PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Bellaire	5
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

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UPDATE

Example: Give all employees in the 'Research' department a 10% raise in salary.

The modified **Salary** value depends on the original **Salary** value in each tuple

- **Salary** on the right of = refers to the old **Salary** value before modification
- **Salary** on the left of = refers to the new **Salary** value after modification

```
UPDATE EMPLOYEE
SET Salary = Salary*1.1
WHERE Dno IN
(SELECT Dnumber FROM DEPARTMENT
WHERE Dname='Research');
```

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Examples of Failures

Operation:

INSERT INTO EMPLOYEE

VALUES ('Cecilia', 'F', 'Kolonsky', NULL, '1960-04-05', '6357 Windy Lane, Katy, TX', 'F', 28000, NULL, 4);

Result: Error while executing SQL query on database 'COMPANY': NOT NULL constraint failed: EMPLOYEE.Ssn

Operation:

INSERT INTO EMPLOYEE

VALUES ('Cecilia', 'F', 'Kolonsky', '999887777', '1960-04-05', '6357 Windy Lane, Katy, TX', 'F', 28000, '987654321', 4);

Result: Error while executing SQL query on database 'COMPANY': UNIQUE constraint failed: EMPLOYEE.Ssn

Operation:

INSERT INTO EMPLOYEE

VALUES ('Cecilia', 'F', 'Kolonsky', '677678989', '1960-04-05', '6357 Windswept, Katy, TX', 'F', 28000, '987654321', 7);

Result: Error while executing SQL query on database 'COMPANY': FOREIGN KEY constraint failed

EMPLOYEE									
Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT			
Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

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Examples of Failures

Operation:

DELETE FROM EMPLOYEE
WHERE Ssn = '333445555';

Result: Error while executing SQL query on database 'COMPANY': FOREIGN KEY constraint failed.

Operation:

DELETE FROM EMPLOYEE
WHERE Ssn = '999887777';

Result: Error while executing SQL query on database 'COMPANY': FOREIGN KEY constraint failed.

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

WORKS_ON

EsSn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

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Failure Examples

Operation:

UPDATE EMPLOYEE
SET Dno=7 WHERE Ssn = '999887777';

Result: Error while executing SQL query on database 'COMPANY': FOREIGN KEY constraint failed

Operation:

UPDATE EMPLOYEE
SET Ssn = '987654321' WHERE Ssn = '999887777';

Result: Error while executing SQL query on database 'COMPANY': UNIQUE constraint failed: EMPLOYEE.Ssn

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

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